

From A Hacker's Diary

Waiting to *boot* the spirits of Freedom

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Recently, I launched the alpha release of Integle—an Intelligent Search Engine based on artificial intelligence—(www.integle.org), and have many volunteers for the project. In fact, I'm really looking for someone to do the design work, as I want to have a decent lay out (rather than that of Jade) and wanted to implement the libraries of the new language (that may solve the current rendering issues and I need not use PHP to deliver stuff).

Alexandre Oliva of FSF Latin America, is associated with the kernel-libre project of GNU. This project aims to remove all the non-free part of the code. Due to the huge inflow of e-mails that I have been receiving, I inadvertently missed out on a mail from him. I wanted to have a look at the automation scripts that they were using for the process. He gave me all the relevant info that I wanted, and I learned more about the deblob-main that can be used in the deletion-process of non-Free bits from the Linux kernel. The script will, in turn, utilise `deblob-<kver>` and `deblob-check` (you can cross check whether a source file contains any remaining or suspicious firmware).

My motive was something else. I could apply some tools while experimenting with another project of FSF—Coreboot (www.coreboot.org), earlier known as LinuxBIOS. The project is aimed at replacing the standard BIOS firmware, found in most computers, with free software. It has many other features other than being absolutely free software. It boots faster (3 seconds to Linux console!) and is written in C (contains virtually no assembly code). Further, it averts the need for very slow / buggy / proprietary BIOS.

The Coreboot project started in 1999, in the Advanced Computing Laboratory at Los Alamos National Laboratory. I experimented with it on

an AMD Geode LX (x86-compatible system-on-chip microprocessor) and found it to perform better. Similar was the response of some of my friends who dared to have a trial in Intel Pentium II / III (Emulation) configurations.

Coreboot usually loads a Linux kernel, but it can load any other stand-alone ELF executable *bits and pieces*. For example, it performs well with Etherboot, which can boot Linux from a boot server or ADLO for Microsoft Windows 2000 / XP and OpenBSD. Coreboot is supported by motherboard vendors like MSI and Gigabyte (and even Google partly sponsors the venture). They offer Coreboot next to their standard BIOS.

Another related project is the Open Firmware project (or OpenBoot) by Sun Microsystems. It is a standard, defining the interfaces of a computer firmware scheme. It is being used by many giants like Sun, Apple, IBM, and most other non-x86 PCI chipset vendors. The Open Firmware even allows the system to load platform-independent drivers straight from the PCI card. It can be accessed through Forth language (structured imperative stack-based computer programming language) shell interface.

The majority of implementations provided by OpenBIOS rely on a supplementary lower-level firmware for hardware initialisation like Coreboot. LANL, AMD, Coresystems GmbH and Linux Networx, Inc contributed much for these projects. Some vendors helped by providing specifications of the hardware interfaces for some of their latest motherboards.

However, Coreboot may work in desktop PCs, servers, clusters, embedded solutions, STBs / HTPC or thin clients, but not for my laptop. Oops! Still I have to use the buggy BIOS for my debian-powered HP DV2713TU. I have only two ways towards a resolution—wait for Coreboot to come out with a solution, or join the project and contribute—then wait for the stable release. I prefer the latter. ☒

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About the Writer

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